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AN INSAR PHASE FILTERING METHOD BASED ON TRANSFORMER NETWORK

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ABSTRACT

In Interferometric Synthetic Aperture Radar (InSAR) data processing, phase filtering has a great impact on the accuracy of the resulting DEM and is therefore an inevitable step. Recently, convolutional neural networks are applied to InSAR phase filtering and show excellent performance, however, most of these deep learning-based methods do not make full use of the self-similarity of phase map, that is, pixels in different or far regions have a close relationship in values or distribution. In this paper, we propose a phase filtering method based on transformer network, which has the advantage of capturing the long-range dependency or exploiting the global features of the phase map, moreover, the deformable convolution is introduced to our method to further extract the local phase features. Experiments validate the efficiency and effectiveness of the proposed method to InSAR phase filtering.

***Index Terms*—** InSAR, Phase filtering, transformer, deep learning,

1. INTRODUCTION

Due to the characteristics of large-scale observation, high-resolution, and microwave imaging, InSAR has a series of important applications, such as topography mapping and deformation monitoring. Influenced by thermal noise, registration error, complex environments, and other factors, the phase map is unavoidably contaminated by noise, therefore, in the past decades, many filtering methods are investigated by researchers.

The traditional phase filtering can be mainly categorized into spatial domain methods, transform domain methods, and their combination. for spatial domain filters, mainly include the mean filter, median filter, Lee filter[1], these filters are basically designed with different filtering windows, statistical principles, and weighting methods. For transform domain filters, the typical and popular method is Goldstein filter[2], which suppresses noise by weighting the phase spectrum and then performing the inverse transform to obtain the filtered phase map. Hybrid filter with the combination of spatial and transform domain was also explored, of which InSAR-BM3D has outstanding performance with elaborate implementation[3]. In addition to the methods described above, most traditional filtering algorithms require manual parameter tuning, and it is difficult to achieve a balance between speed and accuracy.

Recently, deep learning-based methods are applied to InSAR phase denoising gradually. In [4], a deep convolutional network called DeepInSAR is proposed to solve both phase filtering and coherence estimation problems. In [5], the author designs the PFNet with a scale recurrent strategy and realizes both good results and fast calculation. In [6], a generative model called GenInSAR is proposed, which achieves joint phase filtering and coherence estimation via unsupervised training. In [7], the author constructs the denoising convolutional neural network IPDnCNN to reduce the phase noise. The above methods are almost all based on convolutional neural networks, and the main issue of these methods is that the weight of the trained network is static at inference, which makes the network unable to adaptively compute on the input phase.

Recently, transformer network shows great potential in the field of computer vision. In [8], a transformer network calls Uformer is proposed for image restoration, inspired by it, in this paper, we propose a phase filtering method based on the Uformer network, and the deformable convolution is introduced for further enhancements of local phase features extraction[9]. Experiments on simulated and real data validate the efficiency and effectiveness of the proposed method.

2. METHODOLOGY

In this paper, the modified Uformer network is used to filter the noisy phase map, thanks to the hierarchical phase feature fusion, self-attention mechanism, and the local context phase information enhancement, the transformer can effectively extract rich phase features, adaptively calculate based on the input, in the meanwhile, quickly output the filtered phase.

2.1. Network Architecture

The Uformer is an encoder-decoder network with U-shaped structure, and the basic block of Uformer is Lewin transformer block, which extracts features by self-attention and convolution operations. As is shown in Fig.1, a LeWin

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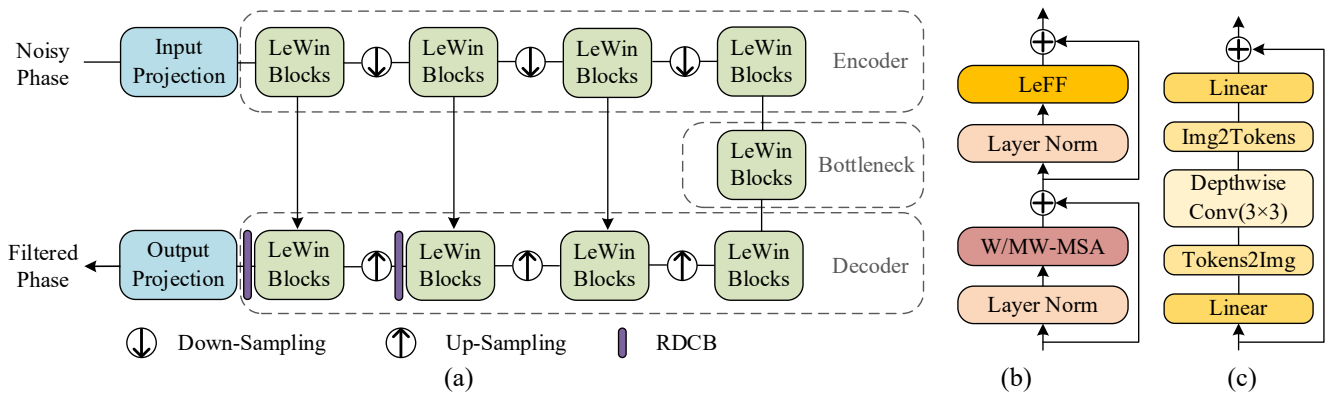

Figure 1. (a) modified Uformer architecture. (b) LeWin transformer block. (c) Locally-enhanced Feed-Forward Network

transformer block is computed as:

$$F_l' = WMSA\left(LN\left(F_{l-1}\right)\right) + F_{l-1}$$
$$F_l = LeFF\left(LN\left(F_l'\right)\right) + F_l' \quad (1)$$

where the WMSA represents the Window-based Multi-head Self-Attention, LN represents the Layer Normalization, LeFF is Locally-enhanced Feed-Forward Network. F_l' and F_l are the output of WMSA module and LeFF module, respectively.

For WMSA module, it performs self-attention in non-overlapping windows on the original feature maps. Specifically, suppose the current feature map is $F \in R^{C\times H\times W}$, the window size is $M\times M$, then in each window, we get flatten and transposed feature $F_w \in R^{M^2\times C}$, if the head number is k, therefore, head dimension is $d = C/k$. The computation of WMSA for one head in non-overlapping windows can be expressed as follows,

$$I = \left\{F^1, F^2, \cdots, F^N\right\}, N = HW/M^2$$
$$Y = Attention\left(F^i W^q, F^i W^k, F^i W^v\right), i = 1, \cdots N \quad (2)$$
$$O = \left\{Y^1, Y^1, \cdots Y^N\right\}$$

where W^q, W^k, W^v represent the project matrices of the queries, keys, and values of one head with the same dimension of $C\times d$. O is the output of the head. Outputs of all heads are concatenated and linearly projected to obtain the final result. For Attention operation, it can be depicted by the following formula:

$$Attention\left(Q,K,V\right) = SoftMax\left(QK^T/\sqrt{d} + B\right)V \quad (3)$$

where B is the position bias, which are learnable parameters. In LeFF module, convolution unit is introduced to the Feed-Forward Network (FFN), aiming to help restore the image details. The computation of LeFF module is as follows:

$$x_{out} = x_{in} + L\left(\mathrm{I}2T\left(DC3\left(T2\mathrm{I}\left(L\left(x_{in}\right)\right)\right)\right)\right) \quad (4)$$

where L, $I2T$, $DC3$, $T2I$ represent linear layer, transform from 2D features to tokens, 3×3 depth-wise convolution, and transform from tokens to 2D features, respectively. And x_{in} and x_{out} are the input and output of LeFF module.

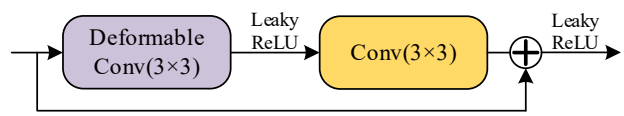

Figure 2. Residual deformation convolution block (RDCB)

Since the shape of interferogram fringes is irregular and various in most cases, the standard convolution may lack the adequate fit over fringes of different shapes due to the rigid receptive field size in the same layer of the network, which possibly lead the over smooth or edge information loss in phase map. Therefore, to better preserve the phase edges, we introduce deformable convolution to the Uformer netwotk[8]. As is shown in Fig.2, similar to ResBlock[10], RDCB (residue deformable convolution block) contains two convolution layers and residue connection, the main difference to ResBlock is that the first layer is replaced by deformable convolution and Leaky ReLU is used here. For the trade-off of capacity and efficiency, we just add RDCB after the last two decode stages as presented in Fig.1.

2.2. Evaluation Metric

To evaluate the performance of phase filtering, the mean square error (MSE) and the number of residues (NOR) are used as the evaluation indexes. MSE is to measure the deviation between the filtered phase and the clean phase, which is defined as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}\left\|p_i - \hat{p}_i\right\|^2 \quad (5)$$

where N is the total number of pixels in a given phase map, p_i and $\hat{p}_i$ are the pixel value in noisy phase map and filtered phase map at coordinate i, respectively.

Residues are the pixels that phase gradient integral not equal to zero in a given closed path[4]. More residues lead to greater errors during the unwrapping process.

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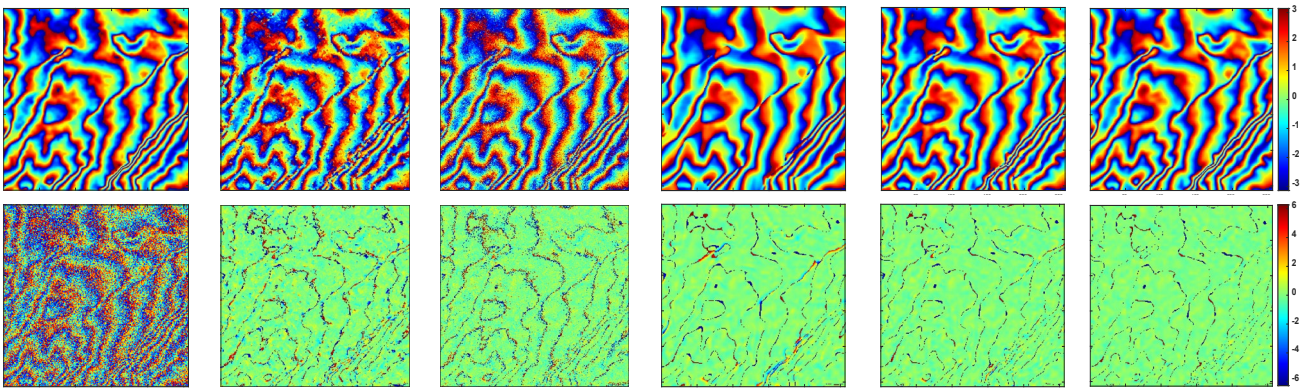

(a) (b) (c) (d) (e)

Figure 3. reference phase (first column) and filtered results of four methods (other columns) on simulated data. (a) clean phase (top)and noisy phase(bottom). (b) - (e) filtered phase(top) and phase difference(bottom) of Lee filter, Goldstein filter, InSAR-BM3D, PFNet and our proposed method respectively

3. EXPERIMENTS

3.1. Datasets

To train and evaluate the proposed transformer network, a dataset is simulated based on ALOS DSM data. First, we crop the DEM data to patches with a size of 256×256. Then, the patches are transformed to absolute phase maps and modulated with 2π to get wrapped phase maps. Last, we add noise in manner of different coherence levels, which ranges from 0.5 to 0.95 with an interval of 0.05. The simulation process refers to the model in the reference[10]. The range of coherence coefficient covers common InSAR data, and phase maps of too low coherence may affect the training of the network. Note that the input of our network is the two-channel image, which contains the real part and imaginary part of noisy phases. We finally produce 8000 pairs of real and imaginary patches to train the network, and 1500 pairs to evaluate the trained model.

3.2. Experiment settings

We train our network by using AdamW optimizer with the momentum terms of (0.9,0.999) and the weight decay of 0.02. The training epochs are set to 50, and the cosine decay strategy is used to decrease the learning rate to 1e-6 with the initial learning rate of 2e-4. The batch size is set to 2. The smooth L1 loss is used as the loss function. Other settings are basically the same as the Uformer[8].

3.3. Experiment results

In Fig.3, The first column shows the clean phase and noisy phase with coherence 0.5, and the second to last column show the filtered results of the Lee filter, Goldstein filter, InSAR-BM3D, PFNet, and our proposed method respectively. It can be seen that the results of the Lee filter and Goldstein filter perform worse than the InSAR-BM3D, PFNet, and our proposed method. Though the filtered phase of InSAR-BM3D and PFNet are close to the clean phase, the edge information in the dense fringe region is partially lost. For our method, not only the noise is significantly removed, but also the phase edges are well preserved.

Table.1. Evaluation results on simulated data

Method	MSE(rad^2)	NOR	Time(s)
No filtering	3.5506	8399	-
Lee	2.0516	931	10.636
Goldstein	1.7433	1015	0.126
InSAR-BM3D	1.7084	141	5.113
PFNet	1.1760	248	0.043
Uformer	0.9414	110	0.065
Proposed	0.9048	62	0.069

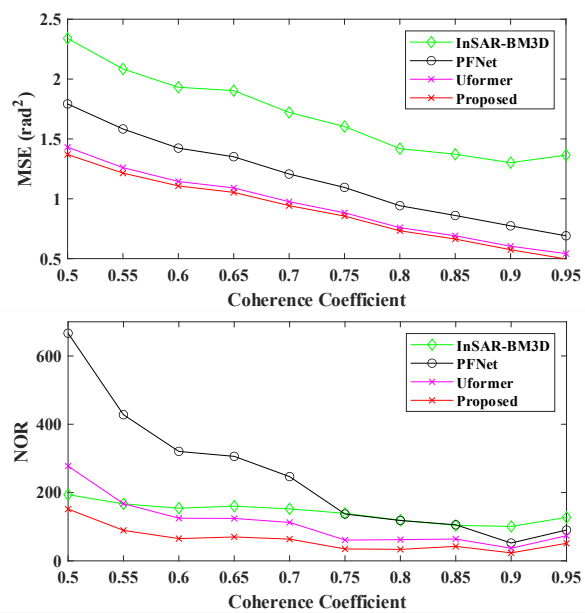

Figure 4. MSE (top) and NOR (bottom) of three methods on simulated data with different coherence coefficients

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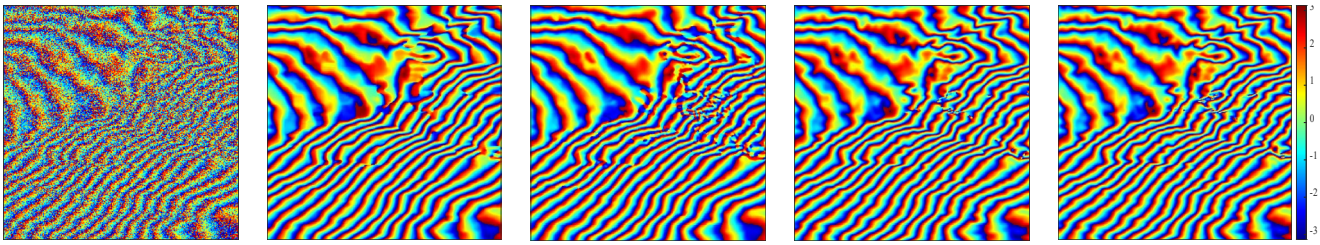

(a) (b) (c) (d) (e)

Figure 5. real data and the filtered results. (a) noisy phase. (b) InSAR-BM3D. (c) PFNet. (d) Uformer. (e) Proposed

To prove the superiority of the network over traditional methods and the effectiveness of our modification to Uformer, evaluation results on 1500 simulated data are listed in table 1. We can see our proposed method outperforms other methods with the lowest MSE and NOR, and the addition of RDCB only brings a little extra time consumption compared to the Uformer.

Results of the InSAR-BM3D, PFNet, Uformer, and our proposed method are close, for a detailed comparison, MSE and NOR index of these methods on simulated data with different coherence are shown in Fig.4. it can be seen that our method always performs better under each coherence.

The filtered results of the three methods on the real data are shown in Fig.5 (data details can be found in [5], and we select regions with both sparse and dense phase fringes for a detailed comparison). It can be seen that our method and Uformer maintains the details of the fringes better, while the resulting fringes of the InSAR-BM3D and PFNet have obvious aliasing and disconnection in some regions.

4. CONCLUSION

In this paper, the phase filtering method based on transformer network is proposed, which is a new attempt of deep learning method on phase filtering. In addition to the advantages of the transformer itself, we introduce the deformable convolution to further enhance the capacity of local phase features extraction. The experimental results show that our method can effectively suppress the phase noise and preserve phase edge information.

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